Optional CFD practice: subcritical airfoil response

Purpose and submission

This open-tool practice exercise prepares you for the in-class CFD interpretation practical. You may work alone or with others and may use the repository documentation, AI tools, notes, and any other resources. You are not being assessed on independent coding.

You may submit the completed evidence sheet for small extra credit and feedback. Feedback addresses whether your claims are supported by your figures and data; it is not a grade for prose or authorship.

What to run

Run `python course/build_week1_subcritical_figures.py` from the repository root. It generates a NACA 2412 geometry, a lift/drag polar from an incidence sweep, and a surface-pressure plot at one subcritical operating point. Do not alter the solver unless you wish to explore beyond the exercise.

Reference relations

For this simplified screening exercise, the two-dimensional airfoil coefficients are treated as representative wing coefficients:

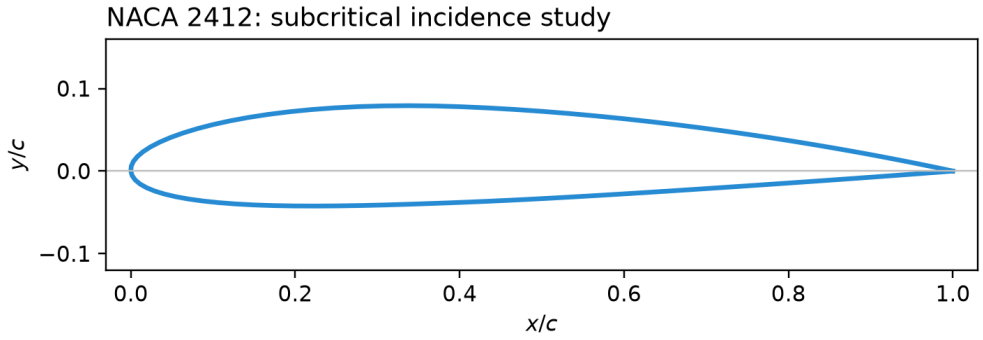
$$L = q_\infty S C_L, \qquad D = q_\infty S C_D, \qquad q_\infty = \tfrac{1}{2}\rho_\infty V_\infty^2.$$

For this practice study, hold $M_\infty = 0.50$ fixed and vary only incidence α. This is intentionally a subcritical study: no shock constraint or transonic design decision is part of Week 1.

Model boundary. This is a two-dimensional inviscid calculation. It is useful for identifying trends in pressure loading and lift, but it does not predict skin-friction drag, separation, three-dimensional effects, structural loads, or aircraft performance.

A. Generate and inspect the incidence sweep

The supplied script evaluates one NACA 2412 section over several angles of attack. Record the output values from your run. If you submit this page, attach the polar and the $\alpha = 2^\circ$ surface-pressure plot.



Incidence α	C_L	C_D	Lift trend	Drag trend	Evidence used	Comment
-2°						
2°						
6°						

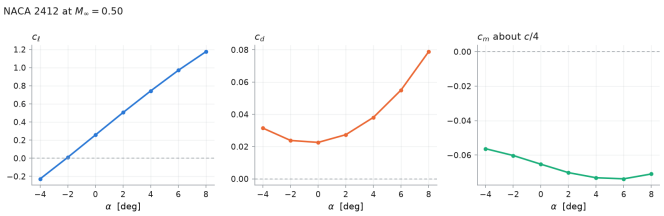
1. On the $\alpha = 2^\circ$ pressure plot, identify the upper and lower surfaces. State which surface has the lower pressure over most of the chord.

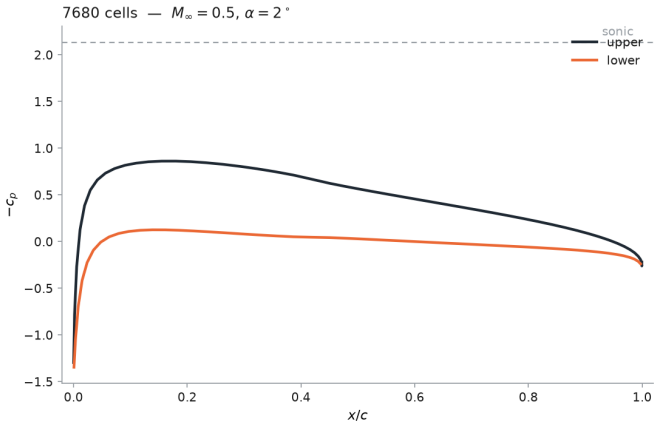
2. Using your polar, describe the trend of C_L as α increases. Do not call the trend "proof" of a real-airfoil performance limit.

3. Identify one conclusion that this inviscid, subcritical exercise cannot support.

B. Compare output types and make a limited selection

The Week 1 figures answer different questions: the polar summarizes integrated loads over incidence, while the C_p curve shows where the pressure loading comes from. This is a response study, not an optimization map.





Screening decision. At this one subcritical operating condition, choose one of the tested incidences for a preliminary operating setting. The setting must satisfy

$$C_L \geq 0.60, \qquad C_D \leq 0.050.$$

If more than one tested incidence satisfies both conditions, select the one with the lower C_D.

4. Select a tested incidence using the stated lift and drag requirements. Cite the two values from the polar that support the decision.

5. Explain why a polar alone cannot tell you where on the airfoil the pressure loading changed.

6. Explain why one C_p curve alone cannot tell you how the integrated lift changes over the full incidence range.

7. Predict what you expect to happen to the upper-surface pressure distribution if α is increased from 2° to 6°. State the observation you would inspect to check that prediction.

Solution key (instructor review)

Values read from the supplied polar are approximate. Award credit for equivalent, figure-supported statements.

1. The black curve is the upper surface and the orange curve is the lower surface. The upper surface has lower pressure over most of the chord: on the plot of $-C_p$, it has the larger value.

2. C_L increases approximately linearly over the plotted incidence range: it is about 0.0 at $-2°$, 0.50 at $2°$, and 0.97 at $6°$. This limited inviscid sweep does not establish a real-airfoil stall or performance limit.

3. Examples: the calculation cannot establish skin-friction drag, separation/stall, three-dimensional wing effects, structural loads, or full-aircraft performance.

4. Select $\alpha = 4°$. From the polar, $C_L \approx 0.74 \geq 0.60$ and $C_D \approx 0.027 \leq 0.050$. The $6°$ point has adequate lift but fails the drag screen ($C_D \approx 0.055$); lower tested incidences fail the lift screen.

5. A polar contains integrated coefficients only. It cannot identify the chordwise location or surface responsible for a pressure-loading change; a $C_p(x/c)$ plot is needed.

6. One C_p curve describes one incidence only. It cannot determine the integrated lift trend over the incidence range; the polar or corresponding curves at multiple incidences are needed.

7. Increasing incidence should strengthen upper-surface suction: the upper-surface C_p becomes more negative (equivalently, its $-C_p$ curve rises) and the pressure difference between surfaces grows. Inspect the upper-surface suction peak and the area between the upper and lower C_p curves.